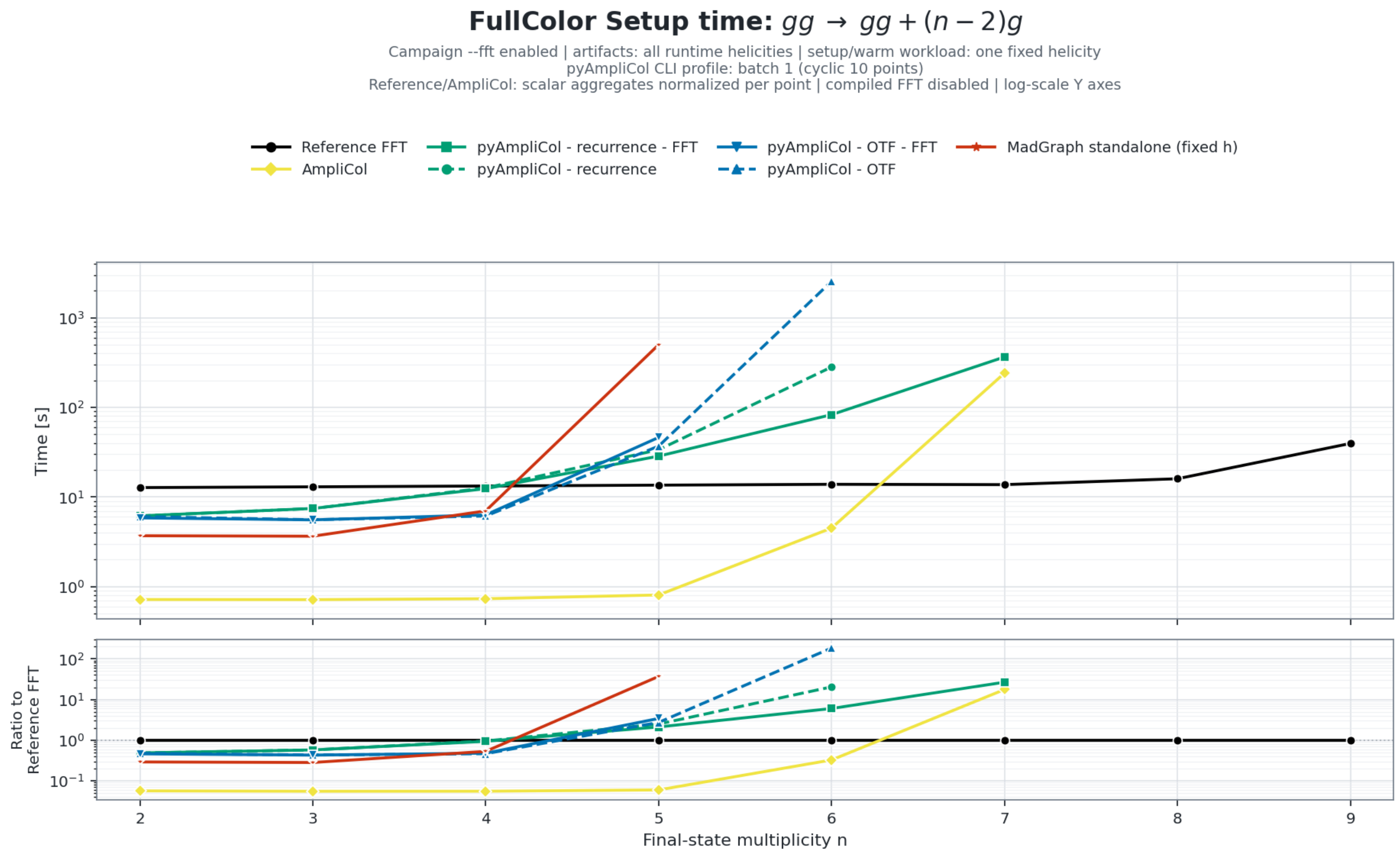
## Current-host fixed-helicity comparison (n=2..9)

IN PROGRESS SNAPSHOT | absent cells are unattempted
--fft | recurrence: persisted all-helicity physical schedule | OTF compact artifact: all runtime helicities
pyAmpliCol setup: generation + fresh load + first fixed-helicity evaluation; OTF includes family warm-up
Reference setup: build/init/first pass | AmpliCol setup: process/color-object generation
Warmed runtime measured separately | compiled FFT disabled
pyAmpliCol CLI profile: batch 1 (cyclic 10 points) | Reference/AmpliCol: scalar aggregates normalized per point




**Failures and omitted points**

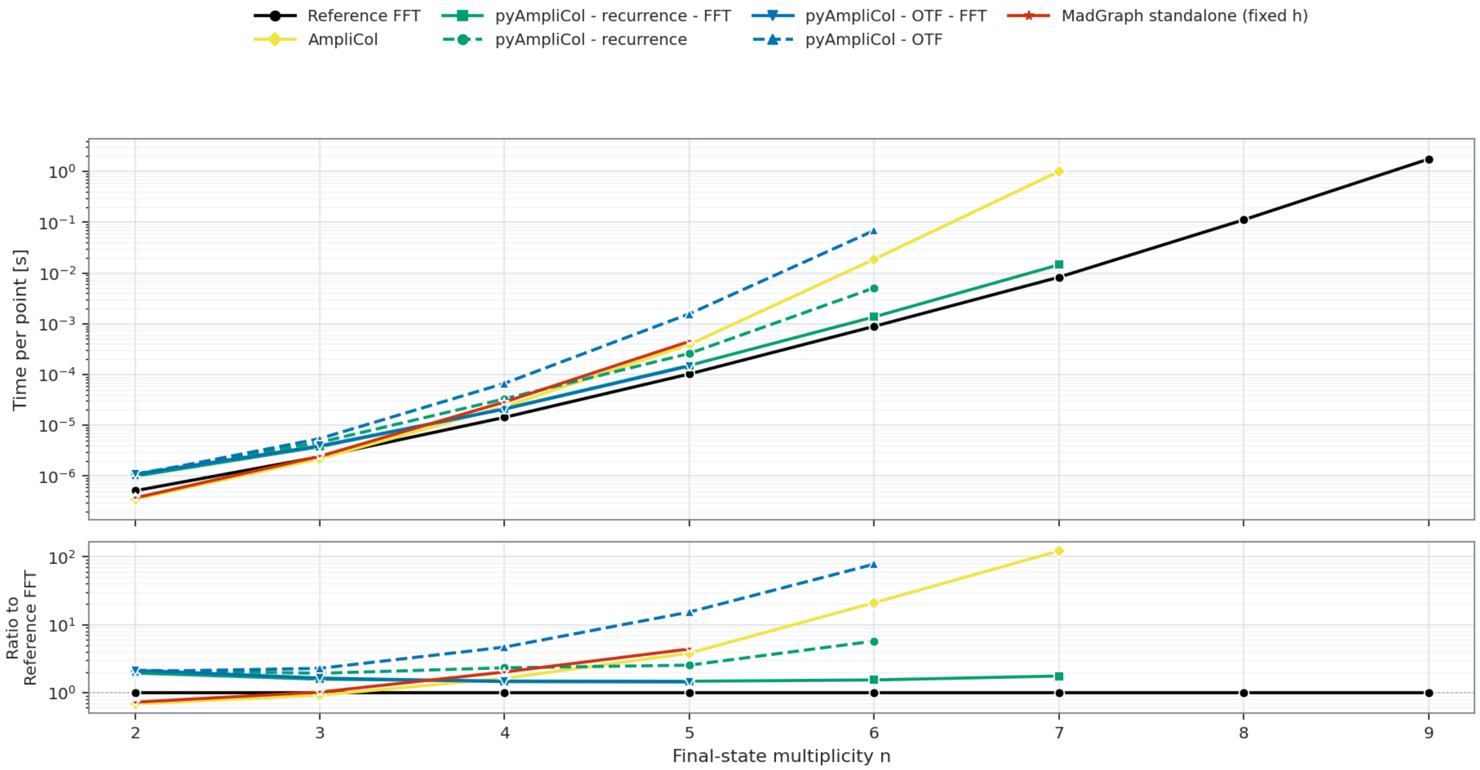
- AmpliCol: n=8 skipped (AmpliCol dense symmetric color-index lower bound is 245.277 GiB, above the 30 GiB cap); n=9 skipped (curve censored after resource limit at n=8).
- pyAmpliCol - recurrence: n=7 failed (memory watchdog limit); n=8-9 skipped (curve censored after resource limit at n=7).
- pyAmpliCol - recurrence - FFT: n=8 failed (memory watchdog limit); n=9 skipped (curve censored after resource limit at n=8).
- pyAmpliCol - OTF - FFT: n=6 failed (60 min runtime limit).
- MadGraph standalone (fixed h): n=6 failed (60 min setup-time limit); n=7-9 N/A (final-plot protocol measures MadGraph only through n=6; higher candidate and reference frontiers are profiled independently).
- Campaign is still running; absent cells are unattempted and omitted.
- Sparse fixed-helicity OTF n=7 extension uses the historical 30-GiB/60-minute limits; gg OTF-FFT n=7 is unattempted after its preserved n=6 runtime frontier.

## Current-host fixed-helicity comparison (n=2..9)

IN PROGRESS SNAPSHOT | absent cells are unattempted
--fft | recurrence: persisted all-helicity physical schedule | OTF compact artifact: all runtime helicities
pyAmpliCol setup: generation + fresh load + first fixed-helicity evaluation; OTF includes family warm-up
Reference setup: build/init/first pass | AmpliCol setup: process/color-object generation
Warmed runtime measured separately | compiled FFT disabled
pyAmpliCol CLI profile: batch 1 (cyclic 10 points) | Reference/AmpliCol: scalar aggregates normalized per point

### FullColor Warmed runtime per point: $gg \to gg + (n-2)g$

Campaign --fft enabled | artifacts: all runtime helicities | setup/warm workload: one fixed helicity
pyAmpliCol CLI profile: batch 1 (cyclic 10 points)
Reference/AmpliCol: scalar aggregates normalized per point | compiled FFT disabled | log-scale Y axes

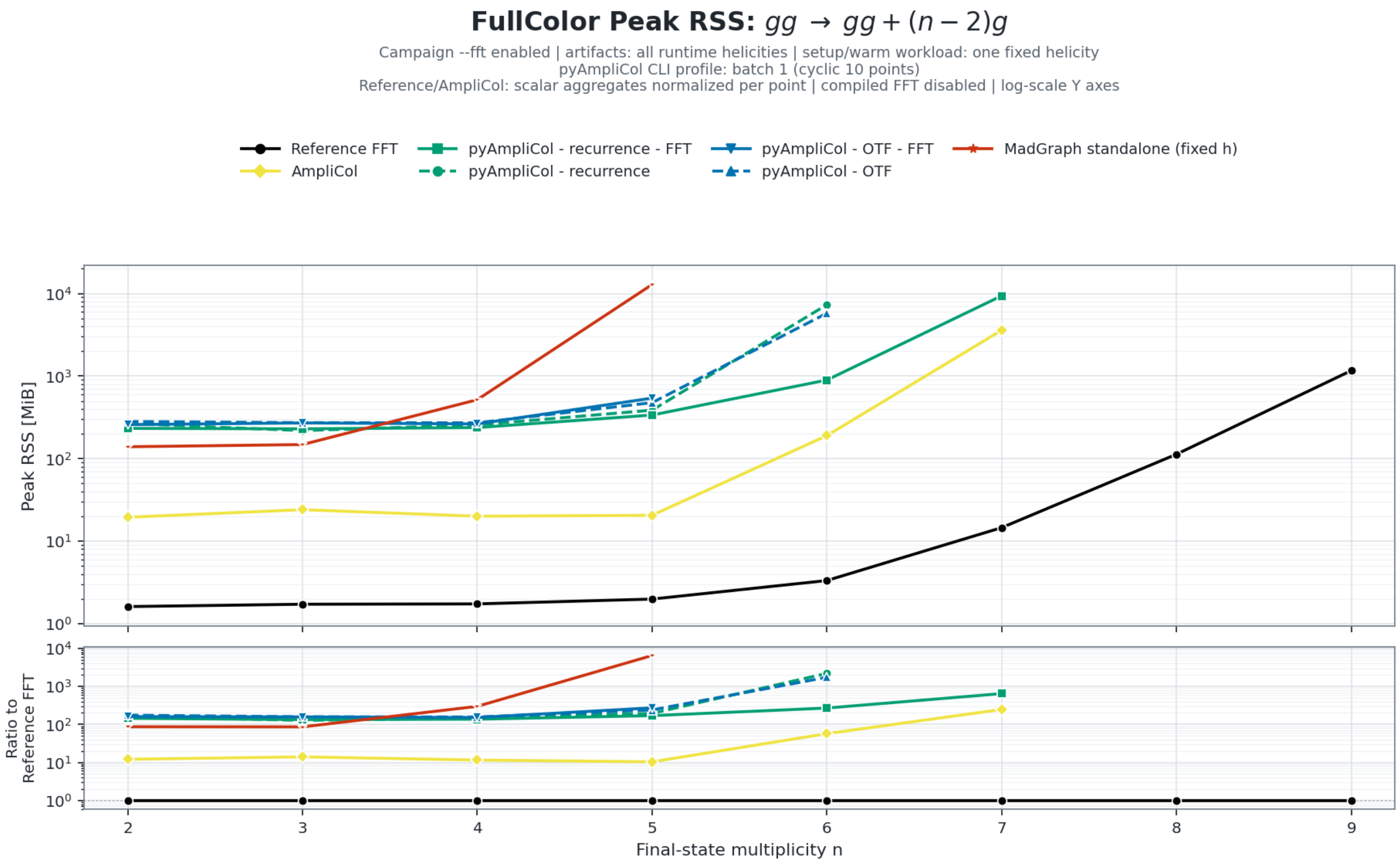



**Failures and omitted points**

- AmpliCol: n=8 skipped (AmpliCol dense symmetric color-index lower bound is 245.277 GiB, above the 30 GiB cap); n=9 skipped (curve censored after resource limit at n=8).
- pyAmpliCol - recurrence: n=7 failed (memory watchdog limit); n=8-9 skipped (curve censored after resource limit at n=7).
- pyAmpliCol - recurrence - FFT: n=8 failed (memory watchdog limit); n=9 skipped (curve censored after resource limit at n=8).
- pyAmpliCol - OTF - FFT: n=6 failed (60 min runtime limit).
- MadGraph standalone (fixed h): n=6 failed (60 min setup-time limit); n=7-9 N/A (final-plot protocol measures MadGraph only through n=6; higher candidate and reference frontiers are profiled independently).
- Campaign is still running; absent cells are unattempted and omitted.
- Sparse fixed-helicity OTF n=7 extension uses the historical 30-GiB/60-minute limits; gg OTF-FFT n=7 is unattempted after its preserved n=6 runtime frontier.

## Current-host fixed-helicity comparison (n=2..9)

IN PROGRESS SNAPSHOT | absent cells are unattempted
--fft | recurrence: persisted all-helicity physical schedule | OTF compact artifact: all runtime helicities
pyAmpliCol setup: generation + fresh load + first fixed-helicity evaluation; OTF includes family warm-up
Reference setup: build/init/first pass | AmpliCol setup: process/color-object generation
Warmed runtime measured separately | compiled FFT disabled
pyAmpliCol CLI profile: batch 1 (cyclic 10 points) | Reference/AmpliCol: scalar aggregates normalized per point

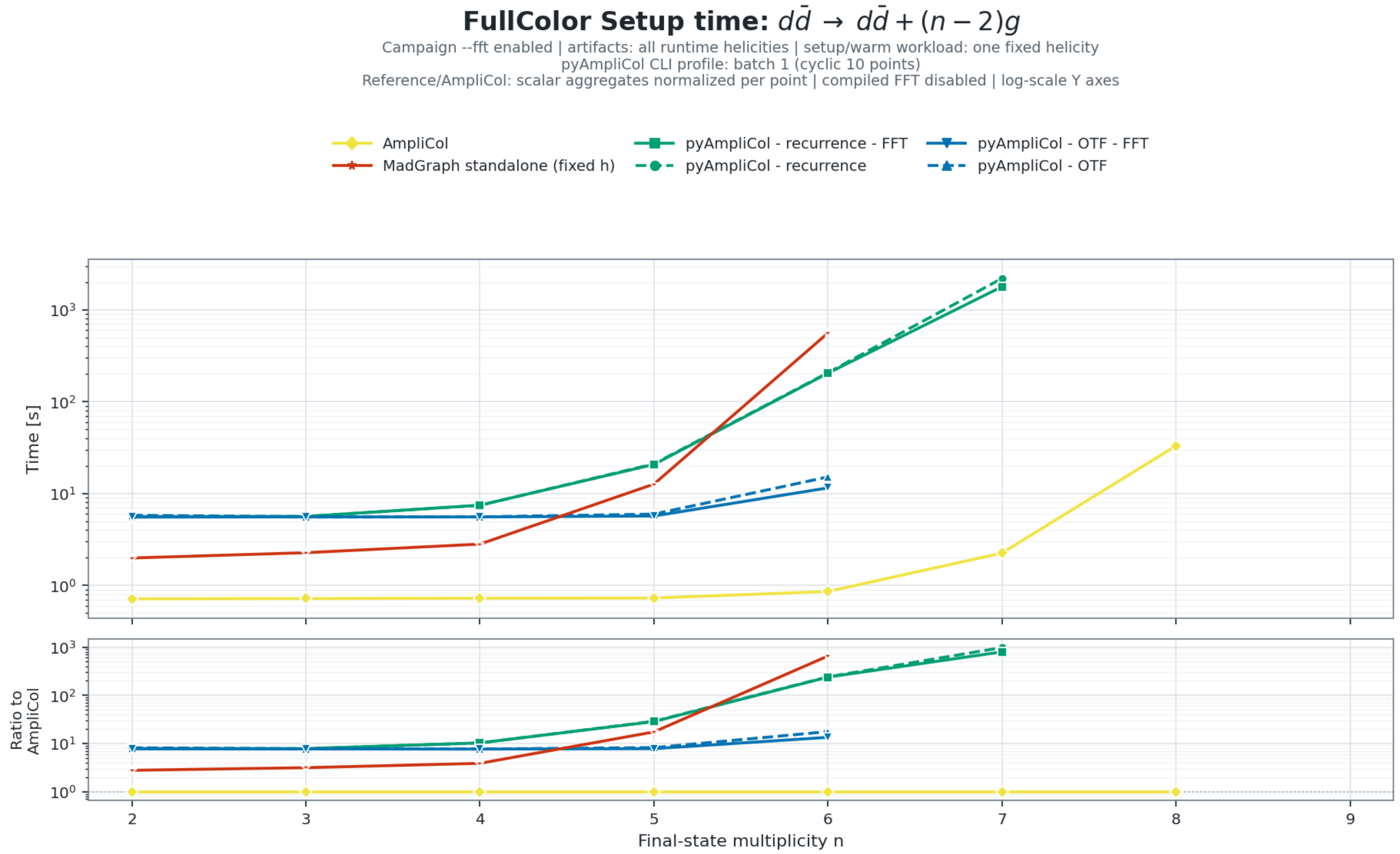



**Failures and omitted points**

- AmpliCol: n=8 skipped (AmpliCol dense symmetric color-index lower bound is 245.277 GiB, above the 30 GiB cap); n=9 skipped (curve censored after resource limit at n=8).
- pyAmpliCol - recurrence: n=7 failed (memory watchdog limit); n=8-9 skipped (curve censored after resource limit at n=7).
- pyAmpliCol - recurrence - FFT: n=8 failed (memory watchdog limit); n=9 skipped (curve censored after resource limit at n=8).
- pyAmpliCol - OTF - FFT: n=6 failed (60 min runtime limit).
- MadGraph standalone (fixed h): n=6 failed (60 min setup-time limit); n=7-9 N/A (final-plot protocol measures MadGraph only through n=6; higher candidate and reference frontiers are profiled independently).
- Campaign is still running; absent cells are unattempted and omitted.
- Sparse fixed-helicity OTF n=7 extension uses the historical 30-GiB/60-minute limits; gg OTF-FFT n=7 is unattempted after its preserved n=6 runtime frontier.

## Current-host fixed-helicity comparison (n=2..9)

IN PROGRESS SNAPSHOT | absent cells are unattempted
--fft | recurrence: persisted all-helicity physical schedule | OTF compact artifact: all runtime helicities
pyAmpliCol setup: generation + fresh load + first fixed-helicity evaluation; OTF includes family warm-up
Reference setup: build/init/first pass | AmpliCol setup: process/color-object generation
Warmed runtime measured separately | compiled FFT disabled
pyAmpliCol CLI profile: batch 1 (cyclic 10 points) | Reference/AmpliCol: scalar aggregates normalized per point

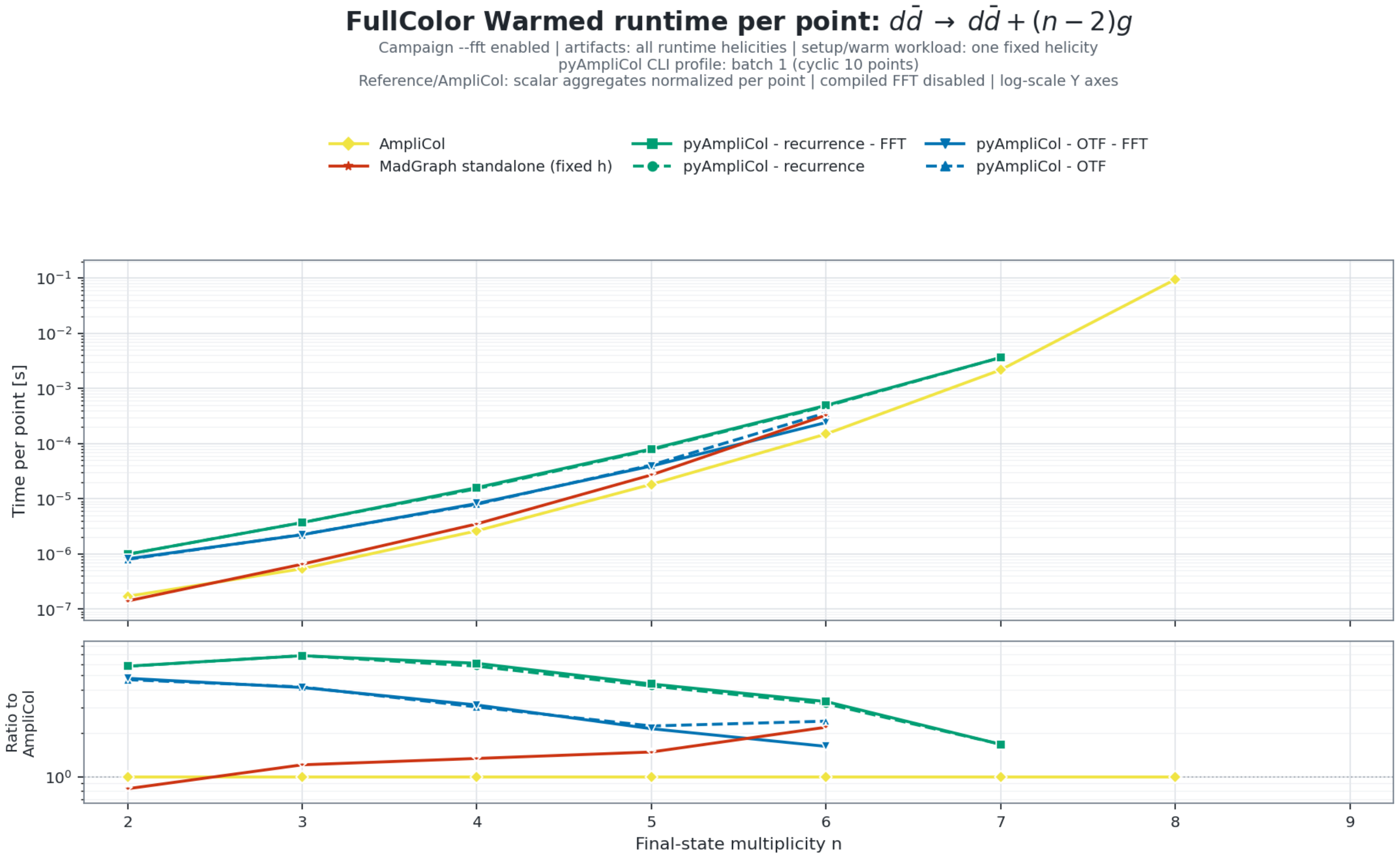



**Failures and omitted points**

- AmpliCol: n=9 failed (structural-memory-limit).
- pyAmpliCol - recurrence: n=8 failed (memory watchdog limit); n=9 skipped (curve censored after resource limit at n=8).
- pyAmpliCol - recurrence - FFT: n=8 failed (memory watchdog limit); n=9 skipped (curve censored after resource limit at n=8).
- MadGraph standalone (fixed h): n=7-9 N/A (final-plot protocol measures MadGraph only through n=6; higher candidate and reference frontiers are profiled independently).
- Campaign is still running; absent cells are unattempted and omitted.
- Sparse fixed-helicity OTF n=7 extension uses the historical 30-GiB/60-minute limits; gg OTF-FFT n=7 is unattempted after its preserved n=6 runtime frontier.
- pyAmpliCol - OTF, pyAmpliCol - OTF - FFT: final publication shown through n=6; higher-n source measurements remain archived and are omitted.

## Current-host fixed-helicity comparison (n=2..9)

IN PROGRESS SNAPSHOT | absent cells are unattempted
--fft | recurrence: persisted all-helicity physical schedule | OTF compact artifact: all runtime helicities
pyAmpliCol setup: generation + fresh load + first fixed-helicity evaluation; OTF includes family warm-up
Reference setup: build/init/first pass | AmpliCol setup: process/color-object generation
Warmed runtime measured separately | compiled FFT disabled
pyAmpliCol CLI profile: batch 1 (cyclic 10 points) | Reference/AmpliCol: scalar aggregates normalized per point

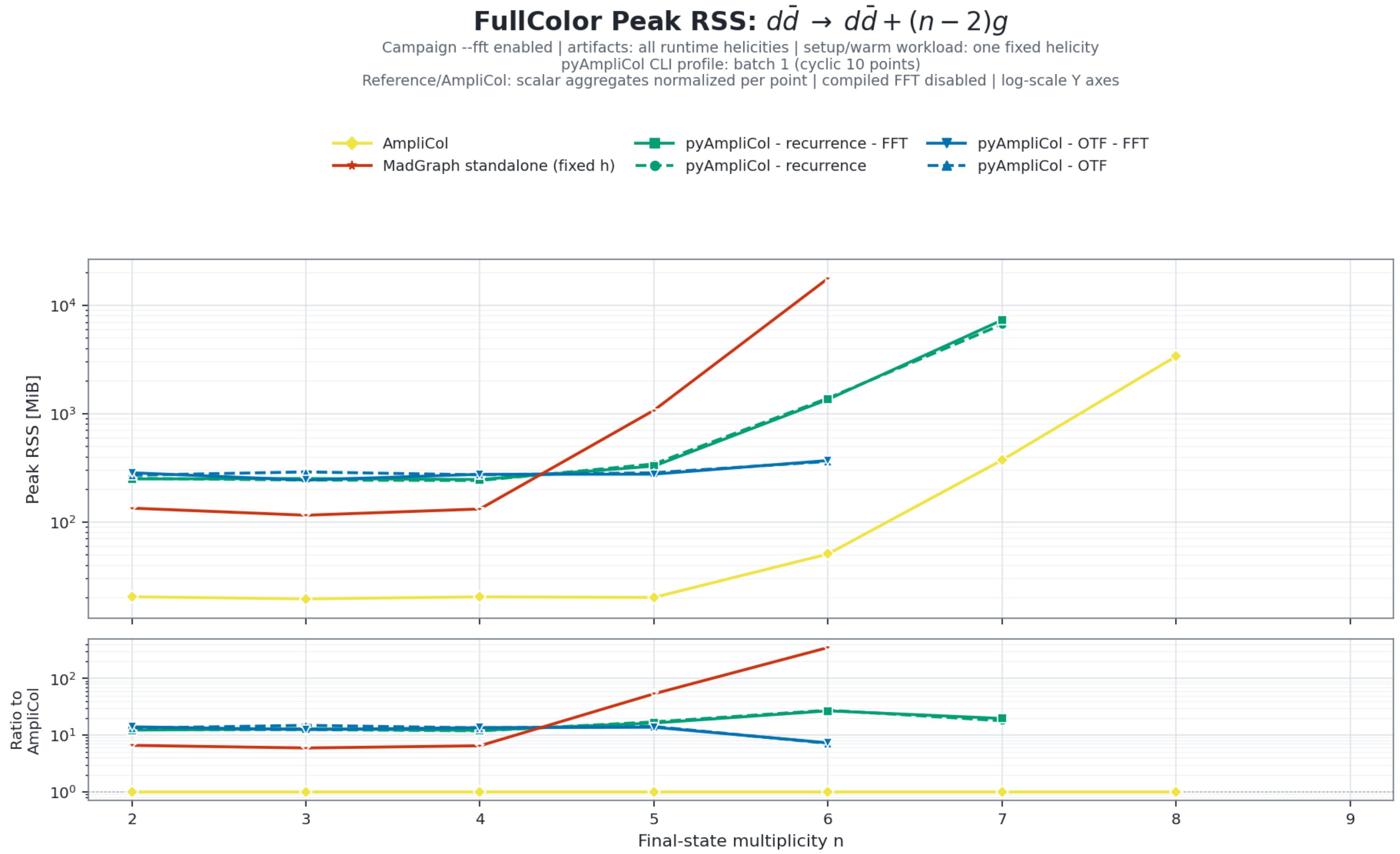



**Failures and omitted points**

- AmpliCol: n=9 failed (structural-memory-limit).
- pyAmpliCol - recurrence: n=8 failed (memory watchdog limit); n=9 skipped (curve censored after resource limit at n=8).
- pyAmpliCol - recurrence - FFT: n=8 failed (memory watchdog limit); n=9 skipped (curve censored after resource limit at n=8).
- MadGraph standalone (fixed h): n=7-9 N/A (final-plot protocol measures MadGraph only through n=6; higher candidate and reference frontiers are profiled independently).
- Campaign is still running; absent cells are unattempted and omitted.
- Sparse fixed-helicity OTF n=7 extension uses the historical 30-GiB/60-minute limits; gg OTF-FFT n=7 is unattempted after its preserved n=6 runtime frontier.
- pyAmpliCol - OTF, pyAmpliCol - OTF - FFT: final publication shown through n=6; higher-n source measurements remain archived and are omitted.

## Current-host fixed-helicity comparison (n=2..9)

IN PROGRESS SNAPSHOT | absent cells are unattempted
--fft | recurrence: persisted all-helicity physical schedule | OTF compact artifact: all runtime helicities
pyAmpliCol setup: generation + fresh load + first fixed-helicity evaluation; OTF includes family warm-up
Reference setup: build/init/first pass | AmpliCol setup: process/color-object generation
Warmed runtime measured separately | compiled FFT disabled
pyAmpliCol CLI profile: batch 1 (cyclic 10 points) | Reference/AmpliCol: scalar aggregates normalized per point




**Failures and omitted points**

- AmpliCol: n=9 failed (structural-memory-limit).
- pyAmpliCol - recurrence: n=8 failed (memory watchdog limit); n=9 skipped (curve censored after resource limit at n=8).
- pyAmpliCol - recurrence - FFT: n=8 failed (memory watchdog limit); n=9 skipped (curve censored after resource limit at n=8).
- MadGraph standalone (fixed h): n=7-9 N/A (final-plot protocol measures MadGraph only through n=6; higher candidate and reference frontiers are profiled independently).
- Campaign is still running; absent cells are unattempted and omitted.
- Sparse fixed-helicity OTF n=7 extension uses the historical 30-GiB/60-minute limits; gg OTF-FFT n=7 is unattempted after its preserved n=6 runtime frontier.
- pyAmpliCol - OTF, pyAmpliCol - OTF - FFT: final publication shown through n=6; higher-n source measurements remain archived and are omitted.